Monte Carlo Simulation under Uncertainty

A Dinner-Deal Decision Rule and the Estimation of π

Excel-based stochastic modeling and Monte Carlo analysis

Portfolio Project Report

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Abstract

This report examines two complementary **Monte Carlo simulation** problems implemented in Excel. The first evaluates a **ratio-based dinner-payment rule** between two salespeople whose weekly sales follow different probability distributions. The second uses random dart locations to estimate a geometric hit probability and, through that probability, **the value of π**.

The projects share a common analytical structure: **uncertain inputs** are represented by probability distributions, problem-specific logic transforms each simulated draw into an outcome, **repeated trials** generate an empirical output distribution, and **statistical summaries** are used to interpret the result.

In the 200-week dinner simulation snapshot, Sam buys dinner 9 times (4.5%), Mary buys 137 times (68.5%), and the check is split 54 times (27.0%), revealing a strongly asymmetric payment pattern under the stated rule and assumptions. In the dart simulation, the estimated values of π for N = 100, 500, 1,000, and 5,000 are 3.3600, 3.2000, 3.2200, and 3.1144, respectively. Although the point estimates do not move monotonically toward π, the 95% confidence-interval width falls from approximately 0.578 at N = 100 to 0.092 at N = 5,000.

The combined analysis illustrates two central uses of **simulation in business analytics: evaluating the distributional consequences of decision rules under uncertainty and quantifying the precision of estimates produced by repeated random experiments**.

Abstract 2
1. Introduction 4
2. Theoretical Framework and Simulation Method 4
2.1 Monte Carlo simulation as input–logic–output analysis 4
2.2 Random-variable generation in Excel 4
2.3 Output analysis, sampling error, and confidence intervals 4
3. Case I: The Dinner Deal 5
3.1 Business context and decision rule 5
3.2 Excel simulation design 5
3.3 Results 5
3.4 Interpretation and business implications 6
4. Case II: Dart Game and Monte Carlo Estimation of π 6
4.1 Problem formulation 6
4.2 Experimental design 7
4.3 Results across sample sizes 7
4.4 Interpretation for simulation practice 8
5. Integrated Discussion 8
6. Managerial and Analytical Implications 9
7. Limitations 9
8. Conclusion 10
References 10

1. Introduction

Many business decisions are made before uncertain outcomes are observed. Sales performance, customer demand, service times, prices, and operational states may vary from period to period, while managers must still specify rules and evaluate their consequences. A deterministic analysis based only on average inputs can conceal how such rules behave across the full range of plausible outcomes. **Monte Carlo simulation** addresses this problem by repeatedly sampling uncertain inputs from specified probability distributions, applying model logic, and analyzing the resulting distribution of outputs.

The project contains two cases designed to illuminate different roles of simulation. **The Dinner Deal** case asks a decision-design question: when two salespeople have different stochastic sales processes and use a fixed performance-ratio rule to determine who pays for dinner, what outcome distribution does that rule generate? **The Dart Game** asks an estimation question: when an event probability can be represented geometrically, how can repeated random sampling estimate that probability and how does simulation precision change as the number of replications increases?

The report is organized as a compact research-style simulation study. It first describes the theoretical framework and Excel implementation, then presents the two case analyses, compares the convergence results across sample sizes, and concludes with managerial implications and limitations. Numerical results are taken from the project workbook; no external data are introduced into the empirical results.

2. Theoretical Framework and Simulation Method

2.1 Monte Carlo simulation as input–logic–output analysis

Monte Carlo simulation relies on **repeated random sampling** and **statistical analysis** to characterize model outputs. **A stochastic model** begins with uncertain inputs, draws realizations from their probability distributions, transforms those realizations through explicit model logic, and aggregates the resulting outputs. This approach is particularly useful when the analyst cares about the distribution of outcomes rather than a single deterministic scenario. Raychaudhuri (2008) describes this workflow as identifying input distributions, generating random variates, processing them through the model, and statistically analyzing the output. Simulation-study guidance likewise emphasizes specifying the data-generating mechanism, varying relevant inputs such as sample size, repeating the experiment, quantifying simulation uncertainty, and reporting results graphically or in tables.

2.2 Random-variable generation in Excel

The workbook implements stochastic inputs directly in Excel. Uniform random draws are generated with **RAND()**. A continuous uniform variable on [a,b] can therefore be generated as a + (b − a)RAND(). Normal random variables are generated through the inverse cumulative distribution function using NORM.INV(RAND(), μ, σ). In both cases, recalculation produces a new simulation realization. Consequently, the displayed frequencies and estimates are stochastic outputs rather than fixed observations.

2.3 Output analysis, sampling error, and confidence intervals

A simulation output should be interpreted with its sampling uncertainty. For a Bernoulli indicator H with simulated sample mean $\hat{p}$, the sample standard deviation s summarizes dispersion of the 0–1 outcomes, while the standard error of the mean decreases with sample size at approximately the square-root rate. The workbook uses the normal-approximation interval

$$\hat{p} \pm 1.96 \times s / \sqrt{N}.$$

For the dart model, $\hat{\pi} = 4\hat{p}$, so the confidence interval for π is obtained by multiplying the probability interval endpoints by four. This distinction between a point estimate and its uncertainty is important: a larger sample does not guarantee that every realized point estimate is closer to the true value, but it systematically reduces the typical sampling variability. Simulation reporting literature similarly recommends reporting uncertainty and graphical summaries rather than presenting only a single mean or point estimate (Wieland & Nelson, 2009).

3. Case I: The Dinner Deal

3.1 Business context and decision rule

Sam and Mary are salespeople whose weekly sales are uncertain. Sam's weekly sales X, measured in thousands of dollars, are modeled as uniformly distributed between 1 and 5:

$$X \sim U[1,5].$$

Mary's weekly sales Y are modeled as Normally distributed with mean 3 and standard deviation 1.3, also in thousands of dollars:

$$Y \sim N(3,\ 1.3^2).$$

Each week, the two salespeople compare their performance using the ratio R = X/Y. The dinner-payment rule is threshold based: Sam buys if R < 0.3; Mary buys if R > 0.7; otherwise, they split the check. The analytical question is not what happens in one week, but what distribution of payment outcomes emerges when the rule is applied repeatedly under the assumed uncertainty.

3.2 Excel simulation design

The workbook simulates 200 weeks. For each week, Sam's sales are generated as 1 + 4×RAND(), Mary's sales are generated with NORM.INV(RAND(),3,1.3), and the ratio X/Y is calculated. The payment outcome is coded as one of three mutually exclusive events: 1 for Sam, 2 for Mary, and 3 for Split. Frequencies are then counted across the 200 simulated weeks. This converts the weekly decision rule into an empirical distribution of long-run consequences.

3.3 Results

Outcome	Frequency	Share of 200 weeks
Sam buys	9	4.5%
Mary buys	137	68.5%
Split	54	27.0%

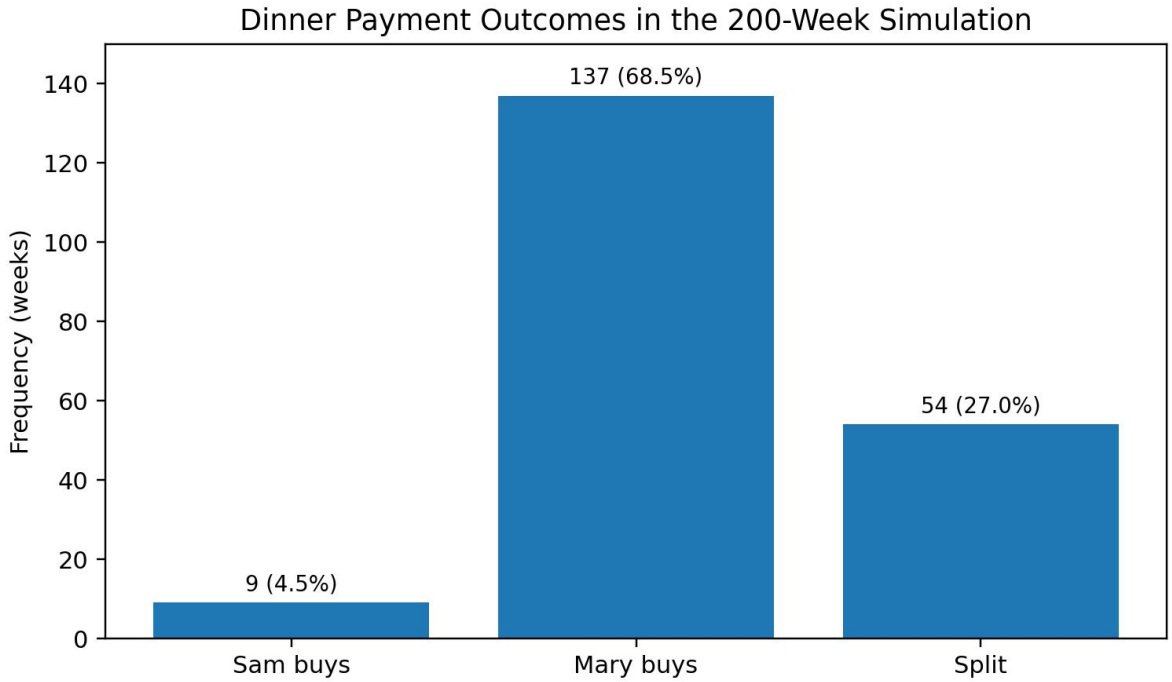


Figure 1. Dinner-payment outcome frequencies in the 200-week simulation snapshot.

The simulated outcome distribution is highly **asymmetric**. Mary pays in more than two-thirds of the simulated weeks, while Sam pays in fewer than one in twenty. Splitting occurs in slightly more than one quarter of the weeks. The difference is not a minor sampling fluctuation in the displayed realization: the observed gap between 137 Mary-pay weeks and 9 Sam-pay weeks is large relative to the 200-week simulation horizon.

3.4 Interpretation and business implications

If fairness is defined as an approximately balanced frequency of bearing the dinner cost, the displayed simulation does not support the rule as balanced under the stated distributions and thresholds. It shows that, conditional on the specified sales distributions and the ratio thresholds, the rule produces a strongly uneven allocation of payment outcomes in this simulation.

The broader business implication concerns threshold-based policies. Organizations routinely use cutoffs in incentive plans, service rules, risk classifications, performance management, and resource allocation. A rule that appears neutral in algebraic form may generate systematically asymmetric outcomes when it interacts with heterogeneous underlying distributions. Simulation provides a practical way to stress-test such policies before implementation by examining the frequency and distribution of consequences rather than relying only on average inputs.

4. Case II: Dart Game and Monte Carlo Estimation of π

4.1 Problem formulation

Consider a circular dart board of radius 1 positioned inside a 2×2 square. A dart is assumed to land uniformly at random within the square. The probability of landing on the circular board equals the ratio of the circle's area to the square's area, $p = \pi/4$. The workbook uses this known relationship as a benchmark for a Monte Carlo experiment: **simulate random dart locations, estimate the hit probability**, and **recover π from $\hat{\pi} = 4\hat{p}$.**

The computational model uses one quarter of the geometry. Each simulated dart is represented by coordinates X and Y independently drawn from U[0,1]. The distance from the origin is

$$d = \sqrt{(X^2 + Y^2)}.$$

A hit indicator H equals 1 when d < 1 and 0 otherwise. Averaging H over N simulated darts gives $\hat{p}$, the estimated hit probability. The estimate of π is then $4\hat{p}$.

4.2 Experimental design

Four simulation sizes are compared: N = 100, 500, 1,000, and 5,000. Each sample size is represented by a separate random simulation in the workbook. The design therefore illustrates how Monte Carlo output changes as the number of replications increases, while also preserving the stochastic nature of each run. For each N, the workbook reports the estimated hit probability, sample standard deviation, and an approximate 95% confidence interval.

4.3 Results across sample sizes

N	$\hat{p}$	$\hat{\pi}$	$\lvert\hat{\pi}-\pi\rvert$	π CI lower	π CI upper	CI width
100	0.8400	3.3600	0.2184	3.0711	3.6489	0.5777
500	0.8000	3.2000	0.0584	3.0596	3.3404	0.2808
1,000	0.8050	3.2200	0.0784	3.1217	3.3183	0.1966
5,000	0.7786	3.1144	0.0272	3.0684	3.1604	0.0921

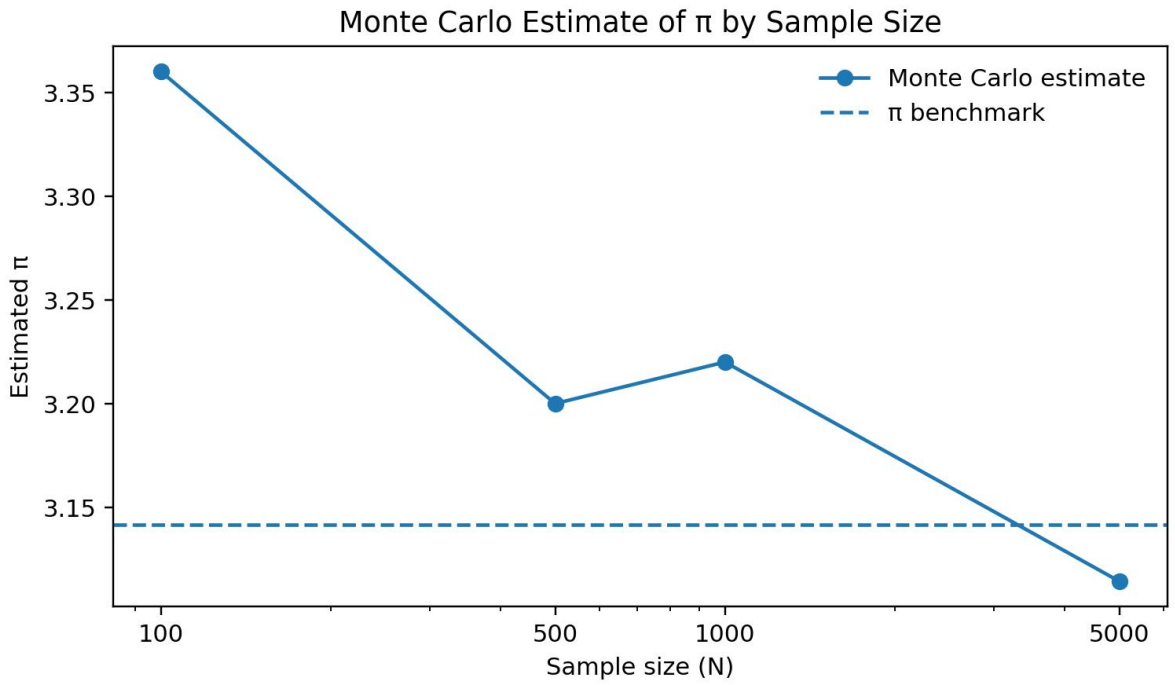


Figure 2. Monte Carlo estimate of π across the four simulation sample sizes.

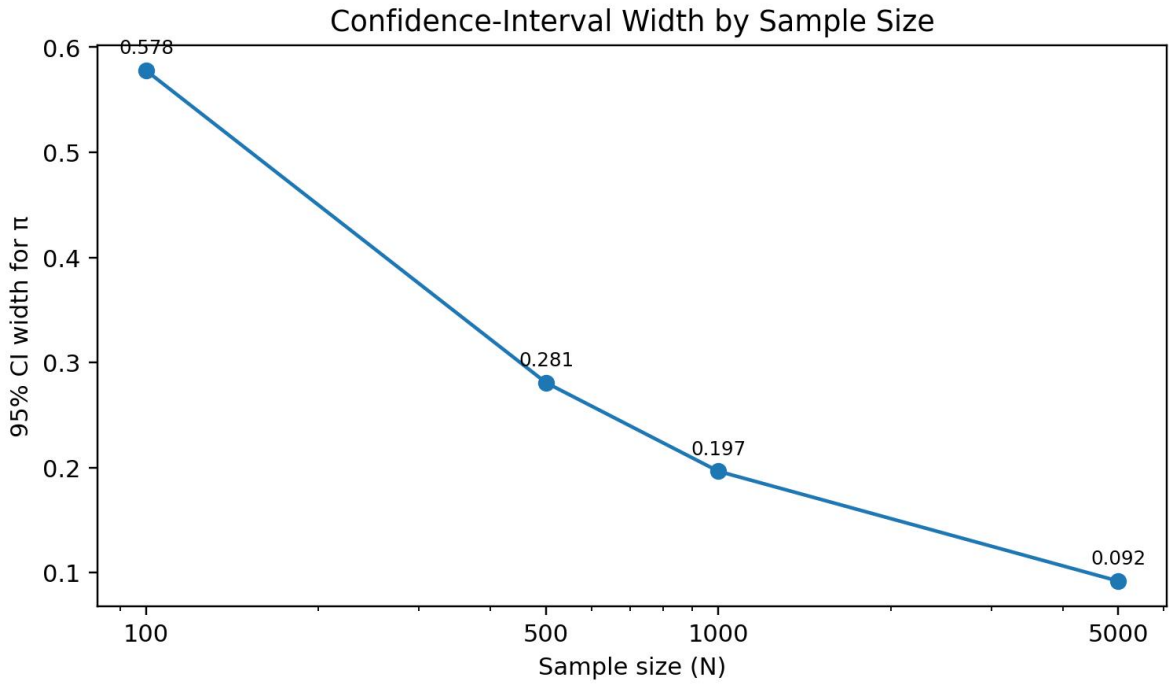


Figure 3. Width of the approximate 95% confidence interval for π by sample size.

The point estimates do not converge monotonically. The N = 500 estimate (3.2000) is closer to π than the N = 1,000 estimate (3.2200), even though the latter uses twice as many trials. This is expected in a stochastic experiment: increasing N reduces sampling variability but does not force every single realized estimate to be closer to the target than every estimate obtained with a smaller sample.

The confidence-interval widths provide a clearer convergence signal. The approximate interval width declines from 0.5777 at N = 100 to 0.2808 at N = 500, 0.1966 at N = 1,000, and 0.0921 at N = 5,000. Relative to N = 100, the N = 5,000 interval is about 84% narrower. This pattern is consistent with the square-root relationship between sample size and standard error: precision improves at a diminishing rate, so large increases in N are required to obtain progressively smaller gains in uncertainty reduction.

4.4 Interpretation for simulation practice

The dart experiment illustrates why **simulation size** is a design choice rather than a cosmetic parameter. A small number of replications may be sufficient for prototyping model logic, but the resulting estimates can be noisy. Larger simulation runs provide more stable evidence when the decision depends on a precise estimate. At the same time, the improvement follows a diminishing-return pattern because standard error decreases with the square root of N. In computationally expensive business models, the analyst therefore faces a practical tradeoff between simulation cost and statistical precision.

5. Integrated Discussion

Although the two cases appear different, they share the same simulation architecture. In the Dinner Deal, uncertain sales are the inputs, the ratio thresholds are the logic, and payment frequencies are the outputs. In the Dart Game, random coordinates are the inputs, the geometric hit rule is the logic, and the estimated probability, π, and confidence interval are the outputs. The common analytical objective is to understand how **uncertainty** propagates through a model.

Simulation component	Dinner Deal	Dart Game
Uncertain inputs	Weekly sales X and Y	Random coordinates X and Y
Probability model	Uniform and Normal	Uniform
Model logic	Ratio thresholds determine payer	Distance rule determines hit
Replication	200 simulated weeks	100–5,000 simulated darts
Primary output	Payment outcome distribution	$\hat{p}$, $\hat{\pi}$, and confidence interval
Decision insight	Rule can create asymmetric burden	Larger N improves precision

The comparison also clarifies two different reasons for using simulation. The Dinner Deal uses simulation to evaluate the distributional consequences of a policy whose output is not obvious from the rule alone. The Dart Game uses simulation as an estimator and makes sampling error visible. In applied business analytics, both functions are important: simulation can reveal the consequences of uncertain systems and can quantify the reliability of estimates used in decisions.

6. Managerial and Analytical Implications

First, decision rules should be evaluated over distributions rather than only at expected values. The Dinner Deal demonstrates that threshold rules can interact with heterogeneous uncertainty to produce strongly uneven outcomes. Similar issues arise in bonus plans, credit rules, service-level classifications, supplier scorecards, and operational triggers.

Second, simulation output should include uncertainty. A point estimate without a confidence interval can create false precision. The Dart Game shows that the reliability of a Monte Carlo estimate depends materially on replication size. For business applications, analysts should select simulation size with reference to the precision required by the decision and the computational cost of additional trials.

Third, transparency in model structure matters. Both models make the path from assumptions to outcomes explicit: distributions define uncertain inputs, formulas encode the logic, and summary statistics describe the outputs. This transparency supports sensitivity analysis and makes it easier to identify whether a surprising result is driven by randomness, model assumptions, or decision rules.

7. Limitations

The Dinner Deal is conditional on its assumed distributions and thresholds. The Normal distribution for Mary's sales has unbounded support and can therefore generate nonpositive values, which are difficult to interpret as weekly sales. Because Y appears in the denominator of X/Y, values close to zero can also generate extreme ratios. The workbook intentionally preserves this model specification; a real managerial application would validate the distributional assumptions against observed sales data and would consider whether truncation or another positive-support distribution is more appropriate.

The fairness interpretation is also definition dependent. This report uses balance in payment frequency as the operational criterion because the simulation records who pays, but other definitions could incorporate dinner cost, sales levels, income, utility, or negotiated preferences. The model does not contain those quantities, so no conclusion about them is warranted.

For the Dart Game, each sample size is represented by one random run rather than many independent replications of the entire estimator. The workbook therefore illustrates convergence and shrinking confidence

intervals but does not empirically estimate the full sampling distribution of $\hat{\pi}$ at each N. In addition, the approximate confidence intervals use the course normal-approximation formula. The geometric benchmark is known analytically, so the case is primarily methodological rather than a situation in which simulation is computationally necessary.

Finally, the workbook uses volatile random-number functions. Recalculation produces new simulation realizations, so exact frequencies and point estimates can change. The numerical results in this report correspond to the workbook snapshot used for the portfolio analysis and should be interpreted as one realization of the specified stochastic models.

8. Conclusion

This project demonstrates an **end-to-end Monte Carlo simulation workflow in Excel: formalizing uncertainty with probability distributions, generating random inputs, encoding problem-specific logic, replicating stochastic experiments, summarizing output distributions, quantifying sampling uncertainty,** and **translating numerical results into decision-oriented interpretation**. The Dinner Deal shows how a simple threshold policy can produce an unexpectedly asymmetric distribution of consequences. The Dart Game shows how repeated random sampling can estimate a probability and how increasing the simulation size improves precision even when point estimates fluctuate non-monotonically.

The combined lesson is that simulation is most useful when it is treated as an analytical experiment rather than as a random-number exercise. Model assumptions must be explicit, output uncertainty must be reported, and conclusions must be tied to the decision question the model is designed to address. These principles make simulation a practical tool for business analytics, operations, risk analysis, and other settings in which decisions must be made under uncertainty.

References

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